

Домашняя работа №1
Приложения двойных и тройных интегралов
Вариант: $18 \bmod 15 = 3$

1.

Дано: $x^3 + y^3 = 2xy$, $x \geq 0$, $y \geq 0$.

Найти: S .

Решение.

$$L : x^3 + y^3 = 2xy, \quad x \geq 0, y \geq 0.$$

$$y = tx \ (t \geq 0) \Rightarrow x^3 + (tx)^3 = 2x \cdot (tx) \Rightarrow x^3(1 + t^3) = 2tx^2 \Rightarrow x = \frac{2t}{1 + t^3}, \quad y = \frac{2t^2}{1 + t^3}.$$

$$\begin{cases} x = x(t) = \frac{2t}{1 + t^3}, \\ y = y(t) = \frac{2t^2}{1 + t^3}, \end{cases} \quad t \in [0, +\infty).$$

$$S = \left| \oint_L y \, dx \right| = \left| \int_0^{+\infty} y(t) x'(t) \, dt \right|.$$

$$x'(t) = \left(\frac{2t}{1 + t^3} \right)' = \frac{2(1 + t^3) - 2t \cdot 3t^2}{(1 + t^3)^2} = \frac{2(1 - 2t^3)}{(1 + t^3)^2}.$$

$$S = \left| \int_0^{+\infty} \frac{2t^2}{1 + t^3} \cdot \frac{2(1 - 2t^3)}{(1 + t^3)^2} \, dt \right| = \left| \int_0^{+\infty} \frac{4t^2(1 - 2t^3)}{(1 + t^3)^3} \, dt \right|.$$

$$u = t^3 \Rightarrow du = 3t^2 \, dt \Rightarrow S = \left| \frac{4}{3} \int_0^{+\infty} \frac{1 - 2u}{(1 + u)^3} \, du \right|.$$

$$w = 1 + u \Rightarrow 1 - 2u = 3 - 2w, \quad w : 1 \rightarrow +\infty.$$

$$\int_0^{+\infty} \frac{1 - 2u}{(1 + u)^3} \, du = \int_1^{+\infty} \left(\frac{3}{w^3} - \frac{2}{w^2} \right) \, dw = \left(-\frac{3}{2w^2} + \frac{2}{w} \right) \Big|_1^{+\infty} = 0 - \left(-\frac{3}{2} + 2 \right) = -\frac{1}{2}.$$

$$S = \left| \frac{4}{3} \cdot \left(-\frac{1}{2} \right) \right| = \frac{2}{3}.$$

Ответ: $S = \frac{2}{3}$.

2.

Дано: $x = 0$, $y = 0$, $z = 0$, $\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$.

Найти: V .

Решение.

$$\Omega : x \geq 0, y \geq 0, z \geq 0, \quad \frac{x}{a} + \frac{y}{b} + \frac{z}{c} \leq 1.$$

$$\frac{z}{c} = 1 - \frac{x}{a} - \frac{y}{b} \Rightarrow z = c \left(1 - \frac{x}{a} - \frac{y}{b} \right), \quad 0 \leq x \leq a, \quad 0 \leq y \leq b \left(1 - \frac{x}{a} \right).$$

$$V = \iiint_{\Omega} 1 dV = \int_0^a \int_0^{b(1-\frac{x}{a})} \int_0^{c(1-\frac{x}{a}-\frac{y}{b})} dz dy dx.$$

$$\int_0^{c(1-\frac{x}{a}-\frac{y}{b})} dz = c \left(1 - \frac{x}{a} - \frac{y}{b} \right).$$

$$V = \int_0^a \int_0^{b(1-\frac{x}{a})} c \left(1 - \frac{x}{a} - \frac{y}{b} \right) dy dx = c \int_0^a \left(\left(1 - \frac{x}{a} \right) y - \frac{y^2}{2b} \right) \Big|_0^{b(1-\frac{x}{a})} dx.$$

$$\left(\left(1 - \frac{x}{a} \right) y - \frac{y^2}{2b} \right) \Big|_0^{b(1-\frac{x}{a})} = \left(1 - \frac{x}{a} \right) b \left(1 - \frac{x}{a} \right) - \frac{b^2(1-\frac{x}{a})^2}{2b} = \frac{b}{2} \left(1 - \frac{x}{a} \right)^2.$$

$$V = c \int_0^a \frac{b}{2} \left(1 - \frac{x}{a} \right)^2 dx.$$

$$u = 1 - \frac{x}{a} \Rightarrow dx = -a du, \quad x : 0 \rightarrow a \Rightarrow u : 1 \rightarrow 0.$$

$$V = \frac{bc}{2} \int_1^0 u^2(-a) du = \frac{abc}{2} \int_0^1 u^2 du = \frac{abc}{2} \left(\frac{u^3}{3} \right) \Big|_0^1 = \frac{abc}{6}.$$

Ответ: $V = \frac{abc}{6}.$

3.

Дано: $z = x^2 + y^2$, $x + y = 5$, $x = 0$, $y = 0$, $z = 0$.

Найти: $(\bar{x}, \bar{y}, \bar{z})$.

Решение.

$$\Omega : 0 \leq z \leq x^2 + y^2, \quad x \geq 0, \quad y \geq 0, \quad x + y \leq 5. \quad D : 0 \leq x \leq 5, \quad 0 \leq y \leq 5 - x.$$

1) Объём.

$$V = \iiint_{\Omega} 1 dV = \iint_D (x^2 + y^2) dx dy = \int_0^5 \int_0^{5-x} (x^2 + y^2) dy dx.$$

$$\int_0^{5-x} (x^2 + y^2) dy = \left(x^2 y + \frac{y^3}{3} \right) \Big|_0^{5-x} = x^2(5-x) + \frac{(5-x)^3}{3}.$$

$$(5-x)^3 = 125 - 75x + 15x^2 - x^3.$$

$$x^2(5-x) + \frac{(5-x)^3}{3} = (5x^2 - x^3) + \left(\frac{125}{3} - 25x + 5x^2 - \frac{x^3}{3} \right) = \frac{125}{3} - 25x + 10x^2 - \frac{4}{3}x^3.$$

$$V = \int_0^5 \left(\frac{125}{3} - 25x + 10x^2 - \frac{4}{3}x^3 \right) dx = \left(\frac{125}{3}x - \frac{25}{2}x^2 + \frac{10}{3}x^3 - \frac{1}{3}x^4 \right) \Big|_0^5.$$

$$V = \frac{625}{3} - \frac{625}{2} + \frac{1250}{3} - \frac{625}{3} = \frac{1250}{3} - \frac{625}{2} = \frac{2500 - 1875}{6} = \frac{625}{6}.$$

2) Координаты $\bar{x}, \bar{y}$.

$$\bar{x} = \frac{1}{V} \iiint_{\Omega} x dV = \frac{1}{V} \iint_D x(x^2 + y^2) dx dy = \frac{1}{V} \int_0^5 \int_0^{5-x} x(x^2 + y^2) dy dx.$$

$$\int_0^{5-x} x(x^2 + y^2) dy = x \left(x^2 y + \frac{y^3}{3} \right) \Big|_0^{5-x} = x \left(x^2(5-x) + \frac{(5-x)^3}{3} \right) = x \left(\frac{125}{3} - 25x + 10x^2 - \frac{4}{3}x^3 \right).$$

$$\iiint_{\Omega} x dV = \int_0^5 \left(\frac{125}{3}x - 25x^2 + 10x^3 - \frac{4}{3}x^4 \right) dx = \left(\frac{125}{6}x^2 - \frac{25}{3}x^3 + \frac{5}{2}x^4 - \frac{4}{15}x^5 \right) \Big|_0^5.$$

$$\iiint_{\Omega} x dV = \frac{3125}{6} - \frac{3125}{3} + \frac{3125}{2} - \frac{2500}{3} = \frac{3125 - 6250 + 9375 - 5000}{6} = \frac{1250}{6} = \frac{625}{3}.$$

$$\bar{x} = \frac{\frac{625}{3}}{\frac{625}{6}} = 2.$$

$$\iiint_{\Omega} y dV = \iint_D y(x^2 + y^2) dx dy \Rightarrow \bar{y} = \bar{x} = 2.$$

3) Координата $\bar{z}$.

$$\bar{z} = \frac{1}{V} \iiint_{\Omega} z dV, \quad \iiint_{\Omega} z dV = \iint_D \left(\int_0^{x^2+y^2} z dz \right) dx dy = \iint_D \frac{(x^2 + y^2)^2}{2} dx dy.$$

$$\iiint_{\Omega} z dV = \int_0^5 \int_0^{5-x} \frac{(x^2 + y^2)^2}{2} dy dx, \quad (x^2 + y^2)^2 = x^4 + 2x^2y^2 + y^4.$$

$$\begin{aligned} \int_0^{5-x} \frac{(x^2 + y^2)^2}{2} dy &= \int_0^{5-x} \left(\frac{x^4}{2} + x^2y^2 + \frac{y^4}{2} \right) dy = \left(\frac{x^4}{2}y + x^2\frac{y^3}{3} + \frac{1}{2}\frac{y^5}{5} \right) \Big|_0^{5-x} \\ &= \frac{x^4}{2}(5-x) + \frac{x^2}{3}(5-x)^3 + \frac{1}{10}(5-x)^5. \end{aligned}$$

$$(5-x)^3 = 125 - 75x + 15x^2 - x^3, \quad (5-x)^5 = 3125 - 3125x + 1250x^2 - 250x^3 + 25x^4 - x^5.$$

$$\frac{x^4}{2}(5-x) = \frac{5}{2}x^4 - \frac{1}{2}x^5,$$

$$\frac{x^2}{3}(5-x)^3 = \frac{x^2}{3}(125 - 75x + 15x^2 - x^3) = \frac{125}{3}x^2 - 25x^3 + 5x^4 - \frac{1}{3}x^5,$$

$$\frac{1}{10}(5-x)^5 = \frac{625}{2} - \frac{625}{2}x + 125x^2 - 25x^3 + \frac{5}{2}x^4 - \frac{1}{10}x^5.$$

$$\int_0^{5-x} \frac{(x^2 + y^2)^2}{2} dy = \frac{625}{2} - \frac{625}{2}x + \left(\frac{125}{3} + 125 \right) x^2 - 50x^3 + \left(\frac{5}{2} + 5 + \frac{5}{2} \right) x^4 - \left(\frac{1}{2} + \frac{1}{3} + \frac{1}{10} \right) x^5,$$

$$\frac{125}{3} + 125 = \frac{500}{3}, \quad \frac{5}{2} + 5 + \frac{5}{2} = 10, \quad \frac{1}{2} + \frac{1}{3} + \frac{1}{10} = \frac{14}{15},$$

$$\int_0^{5-x} \frac{(x^2 + y^2)^2}{2} dy = \frac{625}{2} - \frac{625}{2}x + \frac{500}{3}x^2 - 50x^3 + 10x^4 - \frac{14}{15}x^5.$$

$$\iiint_{\Omega} z dV = \left(\frac{625}{2}x - \frac{625}{4}x^2 + \frac{500}{9}x^3 - \frac{25}{2}x^4 + 2x^5 - \frac{7}{45}x^6 \right) \Big|_0^5.$$

$$\begin{aligned}
\iiint_{\Omega} z \, dV &= \frac{625}{2} \cdot 5 - \frac{625}{4} \cdot 25 + \frac{500}{9} \cdot 125 - \frac{25}{2} \cdot 625 + 2 \cdot 3125 - \frac{7}{45} \cdot 15625 \\
&= \frac{3125}{2} - \frac{15625}{4} + \frac{62500}{9} - \frac{15625}{2} + 6250 - \frac{109375}{45} \\
&= \frac{3125}{2} - \frac{15625}{4} + \frac{62500}{9} - \frac{15625}{2} + 6250 - \frac{21875}{9} \\
&= \frac{56250}{36} - \frac{140625}{36} + \frac{250000}{36} - \frac{281250}{36} + \frac{225000}{36} - \frac{87500}{36} \\
&= \frac{21875}{36}.
\end{aligned}$$

$$\bar{z} = \frac{\frac{21875}{36}}{\frac{625}{6}} = \frac{21875}{36} \cdot \frac{6}{625} = \frac{21875}{625} \cdot \frac{1}{6} = 35 \cdot \frac{1}{6} = \frac{35}{6}.$$

Ответ: $(\bar{x}, \bar{y}, \bar{z}) = \left(2, 2, \frac{35}{6}\right)$.

4.

Дано: $az = x^2 + y^2$, $z = \sqrt{x^2 + y^2}$, $a > 0$.

Найти: V .

Решение.

$$x = r \cos \theta, \quad y = r \sin \theta, \quad dV = r \, dz \, dr \, d\theta, \quad r \geq 0.$$

$$az = r^2 \Rightarrow z = \frac{r^2}{a}, \quad z = \sqrt{r^2} = r.$$

$$\frac{r^2}{a} = r \Rightarrow r = 0 \text{ или } r = a.$$

$$\Omega: 0 \leq \theta \leq 2\pi, \quad 0 \leq r \leq a, \quad \frac{r^2}{a} \leq z \leq r.$$

$$V = \iiint_{\Omega} 1 \, dV = \int_0^{2\pi} \int_0^a \int_{r^2/a}^r r \, dz \, dr \, d\theta.$$

$$\int_{r^2/a}^r r \, dz = rz \Big|_{r^2/a}^r = r \left(r - \frac{r^2}{a} \right) = r^2 - \frac{r^3}{a}.$$

$$\begin{aligned}
V &= \int_0^{2\pi} \int_0^a \left(r^2 - \frac{r^3}{a} \right) dr \, d\theta = \int_0^{2\pi} \left(\frac{r^3}{3} - \frac{r^4}{4a} \right) \Big|_0^a d\theta = \int_0^{2\pi} \left(\frac{a^3}{3} - \frac{a^3}{4} \right) d\theta \\
&= \int_0^{2\pi} \frac{a^3}{12} d\theta = \frac{a^3}{12} \cdot 2\pi = \frac{\pi a^3}{6}.
\end{aligned}$$

Ответ: $V = \frac{\pi a^3}{6}$.